The interpolation law for the critical scaling of the uniform doubling condition

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Abstract

This paper completes the program, carried out in six companion papers, of determining the critical scaling of the uniform doubling condition of [Shmalo, *Ann. Math. Artif. Intell.* (2024)] for independent samples. For the model truncation regions $G_k = B^n \cap \{x_1 \le 0, \dots, x_k \le 0\}$, $0 \le k \le n-2$, whose lowest transversal boundary stratum — the sphere $\{x \in S^{n-1} : x_1 = \cdots = x_k = 0\}$ — has dimension $j = n-1-k$, we prove that the critical scaling factor is exactly

$$\ell_N \asymp \Big(\frac{\log N}{N}\Big)^{2/(2n-j)} = \Big(\frac{\log N}{N}\Big)^{2/(n+1+k)},$$

with both the upper bound (the condition holds with probability $1-1/N$ at this scaling) and the lower bound (it fails with probability $1-o(1)$ below this scaling, logarithm included) established for every $k \le n-2$; for $k \ge n-1$ the conical exponent $\frac{1}{n}$ of the companion papers applies. This proves the interpolation law conjectured in [*The edge exponent of the uniform doubling condition*] at every stratum dimension, in the orthogonal model. The upper bound rests on a mixed-regime section lemma: at depth h below the extreme point of G_k in an arbitrary direction, the level section has volume $\ge c\,h^{(n-1+k)/2}$, proved by splitting the truncated coordinates at the angular threshold $\sqrt{h}$ — steep coordinates contribute a factor h through a fibre box, shallow ones are absorbed into a spherical cap centered on the stratum, of which the all-negative orthant keeps the exact fraction 2^{-k} by reflection symmetry. The lower bound is a second-moment argument over $\asymp (N/\log N)^{j/(2n-j)}$ congruent caps along the stratum.

1 Setting and main theorem

Let $n \ge 2$, let μ be the uniform probability measure on the closed unit ball $K = B^n$, and let $T = \{X_1, \dots, X_N\}$ be i.i.d. samples with empirical measure $\bar{\mu}_N$. For $0 \le k \le n-2$ consider the truncation set $h_k := \{(e_1, 0), \dots, (e_k, 0)\}$ with region

$$G \;=\; G_k \;:=\; \big\{x \in B^n :\; x_1 \le 0, \dots, x_k \le 0\big\},$$

and the uniform doubling condition (UDC) of [2] for slabs truncated by h_k, with constants $(\kappa, \delta, \sigma, \beta, \ell)$, as in the companion papers [3, 4, 5, 6, 7, 8]. The boundary of G_k is stratified by the subsets $S \subseteq \{1, \dots, k\}$ of active constraints; its lowest-dimensional spherical stratum is

$$E \;:=\; \big\{x \in S^{n-1} :\; x_1 = \cdots = x_k = 0\big\} \;\cong\; S^j, \qquad j \;:=\; n-1-k \;\ge\; 1.$$

The region is uniformly thick: with $c^- := -\frac{1}{2\sqrt{k}}(e_1 + \cdots + e_k)$ (for $k \geq 1$; $c^- := 0$ for $k = 0$),

$$B\Big(c^-,\ \tfrac{1}{2\sqrt{k}} \wedge \tfrac{1}{2}\Big)\ \subseteq\ G_k. \tag{1}$$

Theorem 1.1 (Interpolation law, orthogonal model)**.** *Fix $n \geq 2$ and $0 \leq k \leq n-2$, and fix $\kappa > 1$, $\sigma > 0$, $\beta > 0$, $\delta \geq \frac{3}{N}$ where relevant.*

(i) (Upper bound.) *With the N-independent constants of [3] ($\kappa = 151$, $\sigma = 1$, $\delta = \delta(n,k) > 0$, $\beta = 2$, thickness radius $r_0 = \frac{1}{2\sqrt{k}} \wedge \frac{1}{2}$), there are $C(n,k)$ and N_0 such that for $N \geq N_0$, with probability at least $1 - \frac{1}{N}$, the sample satisfies the UDC for the truncation set h_k with scaling factor*

$$\ell_N\ =\ C(n,k)\left(\frac{\log N}{N}\right)^{2/(n+1+k)}.$$

(ii) (Lower bound.) *There are $c(n,k,\kappa) > 0$ such that with probability $1 - o(1)$, the doubling inequality fails for some slab truncated by h_k at every grid scale below $c\,(\log N/N)^{2/(n+1+k)}$; hence the UDC requires $\ell \geq c\,(\log N/N)^{2/(n+1+k)}$.*

Consequently the critical scaling is exactly $\Theta\big((\log N/N)^{2/(2n-j)}\big)$ for every stratum dimension $1 \leq j \leq n-1$, while for truncation geometry with conical points ($j = 0$) it is $\Theta\big((\log N/N)^{1/n}\big)$ by [4, 6]. The interpolation law of [7, Conjecture 1.2] holds at every j in the orthogonal model.

The cases $k = 0, 1$ are the companion results [5, 7, 8]; we treat all $k \leq n-2$ uniformly, and the reader may keep $k = 2$, $n = 4$ in mind. Throughout, $x = (x_1, \ldots, x_k, x^\flat)$ splits coordinates into the truncated ones and the rest, and for a unit vector u we write $u = (u_1, \ldots, u_k, u^\flat)$.

2 Caps at the stratum

Lemma 2.1 (Extreme points)**.** *Let u be a unit vector, $P := \{i \leq k :\ u_i > 0\}$, and let $\hat{u}$ be u with the P-coordinates replaced by 0. If $\hat{u} \neq 0$, then*

$$\max_{x \in G_k} \langle x, u\rangle\ =\ |\hat{u}|,$$

attained exactly at $p := \hat{u}/|\hat{u}| \in G_k$, which satisfies $p_i = 0$ for $i \in P$ and $p_i \leq 0$ for $i \leq k$, $i \notin P$.

Proof. For $x \in G_k$: $\langle x, u\rangle = \sum_{i \in P} u_i x_i + \langle x, \hat{u}\rangle \leq 0 + |x|\,|\hat{u}| \leq |\hat{u}|$, since $u_i > 0$ and $x_i \leq 0$ for $i \in P$; equality forces $x_i = 0$ ($i \in P$), $|x| = 1$ and $x \parallel \hat{u}$. Finally $p \in G_k$ because $\hat{u}_i = u_i \leq 0$ for $i \leq k$, $i \notin P$. □

If $\hat{u} = 0$ (i.e. u is supported on P), the maximum 0 is attained on the entire face stratum $\{x \in G_k :\ x_i = 0,\ i \in P\}$, a set of positive $(n - |P|)$-dimensional volume; this degenerate case only makes the section bounds below easier and is handled separately in Lemma 4.1.

For $p \in E$ define the *stratum direction* and *stratum cap*

$$u_p := \frac{e_1 + \cdots + e_k + p}{\sqrt{k+1}}, \qquad C_p(h) := \Big\{x \in G_k :\ \langle x, u_p\rangle \geq \tfrac{1}{\sqrt{k+1}} - h\Big\}.$$

By Lemma 2.1 (here $P = \{1, \ldots, k\}$, $\hat{u}_p = p/\sqrt{k+1}$), the maximum of $\langle x, u_p\rangle$ over G_k is $\frac{1}{\sqrt{k+1}}$, attained at p.

Lemma 2.2 (Stratum cap volume and footprint)**.** *There are $0 < c_1(n,k) \le c_2(n,k)$ and $h_1(n,k) > 0$ such that for all $p \in E$ and $0 < h \le h_1$,*

$$c_1\, h^{\frac{n+1+k}{2}} \;\le\; \mu\big(C_p(h)\big) \;\le\; c_2\, h^{\frac{n+1+k}{2}}, \qquad C_p(h) \subseteq \{x:\ |x-p| \le c_3(k)\sqrt{h}\,\}.$$

Proof. By the symmetry of G_k under rotations fixing $e_1, \dots, e_k$ we may take $p = e_{k+1}$. Write $x_i = -t_i$ ($t_i \ge 0$) for $i \le k$, let $y := x_{k+1}$, and let $x'' \in \mathbb{R}^{n-k-1}$ collect the remaining coordinates. With $h' := h\sqrt{k+1}$ the cap is

$$\Big\{\, t_i \ge 0,\ |x| \le 1,\ y \ \ge\ 1 - h' + \textstyle\sum_i t_i \,\Big\},$$

and $|x| \le 1$ gives $y \le \sqrt{1 - \sum t_i^2 - |x''|^2} \le 1 - \frac{\sum t_i^2 + |x''|^2}{2}$. *Upper bound:* the constraints force $\sum_i t_i \le h'$ (so each $t_i \le h'$), $|x''|^2 \le 2h'$, and the y-fibre has length at most h'; hence $\mathrm{vol} \le h' \cdot (h')^k \cdot \omega_{n-k-1}(2h')^{\frac{n-k-1}{2}} \le C h^{\frac{n+1+k}{2}}$. *Lower bound:* restrict to $t_i \le \frac{h'}{8k}$ ($i \le k$) and $|x''|^2 \le \frac{h'}{2}$; then $\sum t_i \le \frac{h'}{8}$, $\sum t_i^2 \le \frac{h'^2}{64k} \le \frac{h'}{64}$ (for $h' \le 1$), and the fibre length is at least

$$\Big(1 - \tfrac{\sum t_i^2 + |x''|^2}{2}\Big) - \Big(1 - h' + \sum t_i\Big) \;\ge\; h'\Big(1 - \tfrac{1}{8} - \tfrac{1}{128} - \tfrac{1}{4}\Big) \;\ge\; \tfrac{h'}{2},$$

using $\sum t_i \le \frac{h'}{8}$, $\frac{\sum t_i^2}{2} \le \frac{h'}{128}$ and $\frac{|x''|^2}{2} \le \frac{h'}{4}$; so $\mathrm{vol} \ge \frac{h'}{2}(\frac{h'}{8k})^k \omega_{n-k-1}(\frac{h'}{2})^{\frac{n-k-1}{2}} \ge c\, h^{\frac{n+1+k}{2}}$. *Footprint:* on the cap, $1 \ge y \ge 1 - h'$, $t_i \le h'$, $|x''|^2 \le 2h'$, so $|x-p|^2 \le kh'^2 + h'^2 + 2h' \le c_3^2 h$. □

Lemma 2.3 (Stall)**.** *Fix $\kappa > 1$, $\sigma > 0$, $\delta \ge \frac{3}{N}$, $p \in E$ and $0 < h \le \frac{H}{4}$, $H \le h_1$. If exactly one sample point lies in $C_p(H)$ and it lies in $C_p(h)$, then for every $w \le H/\kappa$ the slab with direction u_p, center that sample point, and truncation set h_k violates the doubling inequality; hence the UDC fails for every $\ell \le H/\kappa^2$.*

Proof. Identical to the companion stall lemmas: for $w' \le H$ the slab is contained in $C_p(h + \frac{w'}{2}) \subseteq C_p(H)$, so the empirical masses of S_w and $S_{\kappa w}$ both equal $\frac{1}{N} < \min\{\delta, (1+\sigma)\frac{1}{N}\}$. □

3 The lower bound

Proof of Theorem 1.1(ii). This is the second-moment scheme of [5, 6, 7] verbatim, with the cap data of Lemma 2.2. Set $q := (4\kappa^2)^{\frac{n+1+k}{2}} \frac{c_2}{c_1}$, $\varepsilon_3 := \frac{j}{4q(2n-j)}$, $\tau_N := \varepsilon_3 \log N$; define h_N by $\mu(C_p(h_N)) = \frac{\tau_N}{N}$ (independent of p by symmetry), so $h_N \asymp (\tau_N/N)^{2/(n+1+k)}$, and $H_N := 4\kappa^2 h_N$, so $\mu(C_p(H_N)) \le \frac{q\tau_N}{N}$. By Lemma 2.2 the caps have footprint $c_3\sqrt{H_N}$, so a maximal $2c_3\sqrt{H_N}$-separated family $p_1, \dots, p_M \in E \cong S^j$ gives pairwise disjoint outer caps with $M \ge c\, H_N^{-j/2} \ge c'(N/\tau_N)^{\frac{j}{n+1+k}}$ — note $\frac{j}{n+1+k} = \frac{j}{2n-j}$. With E_α the event that $C_{p_\alpha}(H_N)$ contains exactly one sample point and that point lies in $C_{p_\alpha}(h_N)$, and $X := \sum_\alpha \mathbf{1}_{E_\alpha}$:

$$p := \mathbb{P}(E_\alpha) = \tau_N\big(1 - \mu(C(H_N))\big)^{N-1} \ge \tau_N N^{-2q\varepsilon_3}, \qquad \mathbb{E}X = Mp \ge c' N^{\frac{j}{2n-j} - 2q\varepsilon_3} \tau_N^{1 - \frac{j}{2n-j}} \ge c' N^{\frac{j}{2(2n-j)}} \to \infty,$$

and the disjoint-cap correlation ratio is $\exp(O(N\mu(C(H_N))^2)) = 1 + o(1)$, so $\mathbb{P}(X = 0) \le (\mathbb{E}X)^{-1} + o(1) = o(1)$ by Chebyshev. On $\{X \ge 1\}$, Lemma 2.3 forces $\ell > H_N/\kappa^2 = 4h_N \ge c(\log N/N)^{2/(n+1+k)}$. □

4 The section lemma

For a unit vector u with $\hat{u} \neq 0$ let $\tau(x) := \langle x, u\rangle$, $\tau_{\max} := |\hat{u}|$ (Lemma 2.1), and $A_u(h) := \mathrm{vol}_{n-1}(G_k \cap \{\tau = \tau_{\max} - h\})$.

Lemma 4.1 (Section lemma)**.** *There are $a(n,k) > 0$ and $h_0(n,k) > 0$ such that for every unit vector u and every $0 < h \le h_0$,*

$$A_u(h) \;\ge\; a(n,k)\, h^{\frac{n-1+k}{2}}.$$

Proof. As in [8], it suffices to prove the cap-mass bound

$$\Lambda_u^{(n,k)}(h) := \mu\big(\{x \in G_k :\ \tau(x) \ge \tau_{\max} - h\}\big) \;\ge\; b(n,k)\, h^{\frac{n+1+k}{2}} \qquad (0 < h \le h_0), \tag{2}$$

for every unit u with $\hat{u} \neq 0$, together with a bulk anchor; the conversion to sections is given at the end. We prove (2) by induction on k; the base case $k = 0$ is the standard spherical cap bound $\Lambda_u^{(n,0)}(h) \ge c(n) h^{\frac{n+1}{2}}$.

Case 1: $|\hat{u}| \ge \frac{1}{2}$. With $p = \hat{u}/|\hat{u}|$, split the truncated indices into

$$\mathcal{A} := \Big\{i \le k :\ u_i \ge \tfrac{\sqrt{h}}{2k}\Big\}, \quad \mathcal{Z} := \Big\{i \le k :\ i \notin \mathcal{A},\ u_i > 0 \ \text{ or } \ |p_i| < 2\sqrt{h}\Big\}, \quad \mathcal{N} := \{1,\dots,k\}\backslash(\mathcal{A}\cup\mathcal{Z}),$$

so that $u_i \le 0$ and $p_i \le -2\sqrt{h}$ for $i \in \mathcal{N}$. Let $\tilde{u}$ be u with the $\mathcal{A}\cup\mathcal{Z}$-coordinates replaced by 0. Every coordinate zeroed in $\tilde{u}$ but present in $\hat{u}$ lies in $\mathcal{Z} \setminus P$ and satisfies $|\hat{u}_i| = |u_i| \le \max\big(\frac{\sqrt{h}}{2k}, 2\sqrt{h}\big) \le 2\sqrt{h}$, whence

$$|\hat{u}|^2 - |\tilde{u}|^2 \;\le\; 4kh, \qquad |\hat{u}| - |\tilde{u}| \;\le\; \frac{4kh}{|\hat{u}|} \;\le\; 8kh,$$

and $|\tilde{u}| \in [\frac{1}{4}, 1]$ for $h \le h_0(k)$. Set $\tilde{p} := \tilde{u}/|\tilde{u}| \in G_k$; then $\tilde{p}_i = 0$ for $i \in \mathcal{A}\cup\mathcal{Z}$, and for $i \in \mathcal{N}$, $|\tilde{p}_i| = |u_i|/|\tilde{u}| \ge |u_i| \ge 2\sqrt{h}\,|\hat{u}| \ge \sqrt{h}$, with $\tilde{p}_i < 0$.

Decompose $x = y\,\tilde{p} + w$, $w \perp \tilde{p}$. Since $\tilde{p} \perp e_i$ for $i \in \mathcal{A}\cup\mathcal{Z}$, the coordinates $w_i := x_i$ ($i \in \mathcal{A}\cup\mathcal{Z}$) are coordinates of the hyperplane $\tilde{p}^{\perp}$; let $w^{\perp}$ denote the remaining component of w. Define the witness box W by

$$w_i \in \Big[-\min\Big(\frac{h}{8k\,u_i}, \frac{\sqrt{h}}{8}\Big), 0\Big]\ (i \in \mathcal{A}), \qquad w_i \in \Big[-\frac{\sqrt{h}}{8\sqrt{k}}, 0\Big]\ (i \in \mathcal{Z}), \qquad |w^{\perp}| \le \frac{\sqrt{h}}{8},$$

$$1 - \frac{h}{4|\tilde{u}|} \;\le\; y \;\le\; \sqrt{1 - |w|^2}.$$

W is nonempty with a long fibre. We have $|w|^2 \le k\cdot\frac{h}{64k} + k\cdot\frac{h}{64k} + \frac{h}{64} \le \frac{3h}{64}$, so $\sqrt{1-|w|^2} \ge 1 - \frac{3h}{64}$, while $\frac{h}{4|\tilde{u}|} \ge \frac{h}{4}$; the y-fibre has length at least $\frac{h}{4} - \frac{3h}{64} \ge \frac{h}{8}$.

$W \subseteq G_k$. For $i \in \mathcal{A}\cup\mathcal{Z}$: $x_i = w_i \le 0$. For $i \in \mathcal{N}$: $x_i = y\tilde{p}_i + (w^{\perp})_i \le (1-h)(-\sqrt{h}) + \frac{\sqrt{h}}{8} < 0$ for $h \le \frac{1}{2}$. And $|x|^2 = y^2 + |w|^2 \le 1$.

W lies in the cap, after rescaling. Since $w \perp \tilde{p}$ and $u - \tilde{u}$ is supported on $\mathcal{A}\cup\mathcal{Z}$,

$$\tau(x) = y\,\langle\tilde{p}, u\rangle + \langle w, u\rangle = y\,|\tilde{u}| + \sum_{i\in\mathcal{A}\cup\mathcal{Z}} u_i w_i,$$

using $\langle \tilde{p}, u\rangle = \langle \tilde{p}, \tilde{u}\rangle = |\tilde{u}|$ and $\langle w, \tilde{u}\rangle = |\tilde{u}|\langle w, \tilde{p}\rangle = 0$. Now $\sum_{\mathcal{A}} u_i|w_i| \le k \cdot \frac{h}{8k} = \frac{h}{8}$, and on $\mathcal{Z}$ every coordinate has $|u_i| \le 2\sqrt{h}$ (either $0 < u_i < \frac{\sqrt{h}}{2k}$, or $u_i \le 0$ with $|u_i| = |p_i|\,|\hat{u}| < 2\sqrt{h}$), so $\sum_{\mathcal{Z}} |u_i||w_i| \le k \cdot 2\sqrt{h} \cdot \frac{\sqrt{h}}{8\sqrt{k}} \le \frac{\sqrt{k}\,h}{4}$. Hence on W,

$$\tau(x) \;\ge\; \Big(1 - \frac{h}{4|\tilde{u}|}\Big)|\tilde{u}| - \frac{h}{8} - \frac{\sqrt{k}\,h}{4} \;\ge\; |\tilde{u}| - \Big(\frac{3}{8} + \frac{\sqrt{k}}{4}\Big)h \;\ge\; |\hat{u}| - 8kh - kh \;\ge\; \tau_{\max} - 9kh,$$

using $\frac{3}{8} + \frac{\sqrt{k}}{4} \le k$ for $k \ge 1$. Replacing h by $h/(9k)$ in the whole construction (which only changes constants), W lies in the cap of depth h.

Volume. Each $\mathcal{A}$-window has length $\min\big(\frac{h}{8ku_i}, \frac{\sqrt{h}}{8}\big) \ge \frac{h}{8k}$ (using $u_i \le 1$ and $h \le k^2$); each $\mathcal{Z}$-window has length $\frac{\sqrt{h}}{8\sqrt{k}}$; the $w^{\perp}$-ball has dimension $n - 1 - |\mathcal{A}| - |\mathcal{Z}|$; the fibre has length $\ge \frac{h}{8}$. Hence

$$\Lambda_u^{(n,k)}(h) \;\ge\; \frac{\mathrm{vol}(W)}{\mathrm{vol}(B^n)} \;\ge\; c(n,k)\,h \cdot h^{|\mathcal{A}|} \cdot h^{\frac{|\mathcal{Z}| + (n-1-|\mathcal{A}|-|\mathcal{Z}|)}{2}} \;=\; c(n,k)\,h^{\frac{n+1+|\mathcal{A}|}{2}} \;\ge\; c(n,k)\,h^{\frac{n+1+k}{2}},$$

proving (2) in Case 1.

Case 2: $|\hat{u}| < \frac{1}{2}$ (induction on k). Then $\sum_{i \in P} u_i^2 = 1 - |\hat{u}|^2 > \frac{3}{4}$, so some $i_0 \in P$ has $u_{i_0} > \frac{\sqrt{3}}{2\sqrt{k}} \ge \frac{1}{2\sqrt{k}}$. Write $x = (x_{i_0}, x')$ and $u = (u_{i_0}, u')$; note $u' \ne 0$ would fail only if $u = e_{i_0}$, in which case $\hat{u} = 0$, excluded. For $s \in [-\frac{h}{2u_{i_0}}, 0]$ — an interval of length $\frac{h}{2u_{i_0}} \ge \frac{h}{2}$ — the slice $G_k \cap \{x_{i_0} = s\}$ contains $\sqrt{1-s^2} \cdot G'$, where G' is the orthogonal model with parameters $(n-1, k-1)$ in the coordinates x' (the slice is exactly the ball of radius $\sqrt{1-s^2}$ intersected with the $k-1$ remaining half-spaces). On the slice, $\tau(x) = u_{i_0}s + \langle x', u'\rangle \ge -\frac{h}{2} + \langle x', u'\rangle$. The maximum of $\langle x', u'\rangle$ over the unscaled slice region equals $|\widehat{u'}| = |\hat{u}|$ (zeroing the positive truncated coordinates of u' gives the same vector $\hat{u}$, supported away from i_0), and scaling by $\sqrt{1-s^2} \ge 1 - h$ changes it by at most $|\hat{u}|h \le h$. Hence the cap of depth h for (G_k, u) contains, over each slice, the image of the cap of depth h_* for $(G', u'/|u'|)$ with $h_* := \frac{h/4}{|u'|} \ge \frac{h}{4}$, provided $u_{i_0}|s| + \text{scaling loss} \le \frac{h}{2} + h \le 2h$ and the slice-cap depth $\frac{h}{4}$ are budgeted within a total depth $4h$; replacing h by $h/4$ throughout, the inclusion holds at depth h. By the induction hypothesis applied to $(n-1, k-1)$ at depth $\min(h_*, h_0)$ — if $h_* > h_0$ use monotonicity $\Lambda(h_*) \ge \Lambda(h_0) \ge b\,h_0^{(n+k-1)/2} \ge b' h^{(n+k-1)/2}$ — and Fubini over s,

$$\Lambda_u^{(n,k)}(h) \;\ge\; c \cdot \frac{h}{2} \cdot b(n-1, k-1)\,h^{\frac{(n-1)+1+(k-1)}{2}} \;=\; c'\,h^{1+\frac{n+k-1}{2}} \;=\; c'\,h^{\frac{n+1+k}{2}}.$$

Conversion to sections. Exactly as in [8]: by Brunn–Minkowski, A_u is unimodal in the depth; the inner ball (1) provides a bulk depth $h^{\dagger} \ge c(k) > 0$ with $A_u(h^{\dagger}) \ge a_1(n,k) > 0$ (its τ-range is an interval of length $2r_0$ inside the range of G_k, whose sections at distance $\le \frac{r_0}{2}$ from its center have radius $\ge \frac{\sqrt{3}}{2} r_0$). For $h \le h_0 \le h^{\dagger}$: if h is below the mode of A_u, then A_u is nondecreasing on $[0, h]$ and $A_u(h) \ge \Lambda_u(h)/h \ge b\,h^{\frac{n-1+k}{2}}$ by (2); if h is above the mode, unimodality gives $A_u(h) \ge A_u(h^{\dagger}) \ge a_1 \ge a_1 h^{\frac{n-1+k}{2}}$. □

Remark 4.2. *The witness box exploits the key orthogonality $\langle w, \tilde{u}\rangle = 0$: once the shallow coordinates are zeroed into the anchor direction $\tilde{p}$, the level constraint involves only the $\mathcal{A} \cup \mathcal{Z}$ coordinates, so all remaining directions are free at scale $\sqrt{h}$ and the volume count telescopes. The induction of Case 2 is the mechanism by which face strata inherit the law from lower-dimensional models, one truncation per dimension.*

5 Slab bound and proof of the upper bound

Theorem 5.1 (Slab volume)**.** *There are $c(n,k) > 0$, $L_0(n,k) > 0$ with*

$$\mathrm{vol}\Big(G_k \cap \{|\langle y - x, u\rangle| \le L\}\Big) \;\ge\; c\, L^{\frac{n+1+k}{2}} \qquad (x \in G_k,\ u \in S^{n-1},\ 0 < L \le L_0).$$

Proof. Anchor–unimodality as in [4, 8]: anchors at depth $\frac{L}{4}$ from each u-extreme with value $a(n,k)(\frac{L}{4})^{\frac{n-1+k}{2}}$ by Lemma 4.1 applied to u and $-u$; unimodality spreads the minimum anchor value over the middle of the range, whose complement has length $\frac{L}{2}$; the range has length at least $2r_0 \ge 2L$ by (1); and the slab retains length $\ge L$ in the range, of which $\ge \frac{L}{2}$ in the middle. □

Proof of Theorem 1.1(i). Insert Theorem 5.1 into the proof skeleton of [3, 4]: the population dichotomy holds with $r_0 = \frac{1}{2\sqrt{k}} \wedge \frac{1}{2}$ by (1); the VC relative deviation bounds apply to slabs with k truncations; and a sample point in S_w yields $\mu(S_{\kappa w}) \ge 32 u_N$ for all $w \ge \ell_N := C(n,k)(u_N)^{2/(n+1+k)} \asymp (\log N/N)^{2/(n+1+k)}$. □

6 Concluding remarks

The orthogonal model is the normal form of transversal stratified truncation: for half-spaces in general position meeting ∂K and each other at dihedral angles bounded below, the same two-step scheme applies with constants depending on the angles, the witness-box and reflection arguments being replaced by their affine images. The UDC constants themselves are stable under a fixed nonsingular affine change of variables: by the affine-transfer lemma of the high-probability companion paper [3], an affine map with singular values in $[\lambda, \Lambda]$ sends constants $(\kappa, \delta, \sigma, \beta, \ell)$ to $(\kappa^2, \delta, \sigma, \lambda\beta/\kappa, \Lambda\ell)$ in the grid form, and to $(\kappa, \delta, \sigma, \lambda\beta, \Lambda\ell)$ in the all-width form. Thus extending the interpolation law beyond the orthogonal model reduces to the local section-volume estimates, not to a new doubling-formalism issue. Within the model, the program of the seven papers is complete: for every stratum dimension $j \in \{0, 1, \dots, n-1\}$ the critical scaling of the uniform doubling condition is

$$\ell_N \;\asymp\; \Big(\frac{\log N}{N}\Big)^{2/(2n-j)} \quad (1 \le j \le n-1), \qquad \ell_N \;\asymp\; \Big(\frac{\log N}{N}\Big)^{1/n} \quad (j = 0),$$

exponents and logarithms exact. In the language of the stability theory of [1, 2]: the finest resolution at which i.i.d. data can certify the doubling condition is set by the most isolated extremal sample point of the worst boundary stratum of the activation geometry, and degrades gracefully — through the explicit exponent ladder $\frac{2}{n+1}, \frac{2}{n+2}, \dots, \frac{1}{n}$ — as the truncation geometry sharpens from smooth to conical.

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Declaration on generative AI

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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